

REZA JARI
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Technical Skills:

- **Programming:** Python, Java, C, **MIPS32 Assembly (QtMIPS)**, MATLAB
- **ML/AI:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas; CNN/ANN, PCA, **PLS/PLS-DA**, SVM/SVM-DA, cross-validation, hyperparameter search
- **Data Science & Signal Processing (medical/FTIR):** Baseline correction, smoothing/denoising, normalization, feature engineering, **Monte Carlo** simulations, ROC/AUC, calibration/validation protocols
- **XR/HCI & Behavioral Analytics:** **Spatial data** analysis in VR, **body-language**/pose cues, **talk-time** and conversation metrics (Bloom levels, time-to-clarity), **tactile keyboard** interaction logging and patterning
- **LLM Pipelines & Evaluation:** Multi-model orchestration (LLM + STT/TTS), prompt/runtime profiling, **quantization awareness** (8-bit/4-bit, AWQ/GPTQ concepts), dataset logging for conversational evaluation
- **Cloud & Deployment:** AWS (EC2, S3, IAM, VPC basics), **Google Cloud** (Console, Compute Engine, IAM), **Firebase** (Authentication, Firestore/Storage, security rules), **Expo** (EAS) for cross-platform app deployment
- **MLOps & Orchestration:** REST/WebSocket services, job queues, telemetry/logging, **Docker**; model routing & versioning; reproducible notebooks & artifacts
- **Security & Data Architecture:** **Hashed/encrypted identifiers**, encryption in transit/at rest, **multi-tenant** workspace↔user relations, role-based access (RBAC), audit trails
- **Databases & APIs:** SQL (PostgreSQL/MySQL), schema design, migrations; RESTful APIs; JSON/CSV pipelines
- **Dev Tools:** Git, Linux, Jupyter
- **Languages:** English (fluent); Persian/Farsi (fluent)

PROFESSIONAL EXPERIENCE:

Research Assistant - Dr. Sanjay Goel

University at Albany, Albany, NY, USA

May 2026-Present

- **Real-time assistive teleprompter on Meta smart glasses (iOS):** Building an on-device speech-recognition and live-captioning pipeline that streams transcription to Meta smart glasses so users with speech, hearing, and motor-related disabilities can follow and take over a conversation as it happens; architected for edge inference rather than server round-trips to remove network latency and server load from the critical path.
- **Adaptive cyber-deception framework (honeypots and attacker psychology):** Developing a honeypot framework that models attacker cognitive biases and decision heuristics, turning the same psychological levers attackers use against defenders back against them; mines honeypot session and log data to adaptively generate deceptive artifacts that draw intruders deeper into the environment, extending dwell time and intelligence yield.
- **AI/VR employability training framework for adults with cognitive disabilities:** Designing a dynamic, AI-driven VR training framework that adapts task difficulty and scaffolding to the individual learner, targeting the specific workplace-skill barriers that limit employment outcomes; roadmap extends to real-time AR assistance delivered during live on-the-job task performance.

Research Assistant - Dr. Spyros Gallis's Lab

University at Albany, Albany, NY, USA

March 2026-June 2026

- **Modular quantum instrument-control platform:** Designing and building an extensible software platform that unifies control of heterogeneous laboratory instruments used in quantum and nanophotonics experiments, with a driver-abstraction layer that lets new hardware be integrated without changes to the core system.
- **Research-automation workflows:** Implemented configurable experiment workflows that sequence instrument operations, acquisition, and logging into repeatable automated runs, reducing manual operation and improving run-to-run consistency.
- **Architecture & extensibility:** Built the platform around modular, interface-driven components so that control, acquisition, and analysis stages can be extended or swapped independently; designed toward generalizable use across different experimental setups and automated analysis against preseeded reference data.

Research Assistant - Dr. Dumas's Human-Centered Intelligence XR Lab

University at Albany, Albany, NY, USA

October 2024-May 2026

- **Led VR+LLM pipeline** (STT → LLM → TTS) with evaluation queue, telemetry, and quantization-aware configs (8/4-bit).
- **Designed privacy-preserving HCI/learning analytics** (talk-time, Bloom, time-to-clarity); exports to pandas/SPSS/SAS.
- **Built behavioral-biometrics** prototypes (keystroke/mouse/window-focus) and multi-user VR dynamics analytics (spatial/pose).
- **Data governance:** hashed/encrypted identifiers, RBAC, IRB-compliant workflows; contributed figures & methods to papers.
- **CREST - immersive remote workspace (NYS Mesonet):** Built a proof-of-concept immersive remote workspace enabling distributed monitoring and collaboration over NYS Mesonet operations; ran feedback sessions with Mesonet staff and iterated the interaction design and workspace layout against their operational requirements.

Research Assistant - Dr. Igor Lednev's Lab

University at Albany, Albany, NY, USA

April 2023-May 2026

- **ATR-FTIR + ML for early Sjögren's screening:** Built end-to-end Python workflows from raw saliva spectra to validated, reproducible classifiers; supported ongoing manuscript/poster preparation.
- **Preprocessing automation:** Implemented baseline correction (AsLS/IModPoly-style), smoothing/denoising, vector/area normalization, scan aggregation, and outlier screening; created batchable CLI/config-driven runs, reducing manual analyst time per experiment by ~90%.
- **Leakage-safe evaluation:** Enforced strict subject/sample-wise (group-based) splitting, repeatable seeds, and stratified resampling; added leakage checks and systematic ablations to quantify the impact of preprocessing and spectral-range choices.
- **Feature engineering & dimensionality:** Performed band/region selection, k-best selection, derivative spectra (1st/2nd), PCA/PLS component sweeps with variance targets; compared performance across spectral windows and parameter settings.
- **Modeling & strategy evolution:** Benchmarked classical models (PLS-DA, SVM-DA, logistic regression, random forest, k-NN/LDA as needed) and explored CNN baselines; advanced to structured multi-class approaches including hierarchical 3-class classification (N vs X, then R vs S) and one-vs-one (OVO) training.
- **Validation & reporting:** Reported balanced accuracy, macro-F1, ROC/AUC, and clinically relevant sensitivity/specificity operating points where applicable; generated confusion matrices, stability summaries, and run-to-run confidence estimates.

- **Results packaging & communication:** Produced reproducible notebooks and automated figure generation for publication-quality plots; drafted methods/results text, prepared tables/figures, and maintained experiment logs for rapid iteration.
- **Data governance:** Applied de-identification practices (hashed IDs, controlled access), maintained IRB-aligned handling procedures for saliva spectra and associated metadata.

Peer Tutor - Learning Commons

University at Albany, Albany, NY, USA

January 2024-May 2026

- **Tutored students** in data structures, discrete mathematics, biology, chemistry, and calculus I.
- **Improved student performance** and comprehension by breaking down complex topics and teaching practical learning strategies.
- **Supported diverse academic needs** through tailored sessions across STEM subjects.

Research Assistant, Environmental & Sustainable Engineering, Dr. Aynul Bari

University at Albany, Albany, NY, USA

January 2025-May 2025

- **NYS air-quality analysis:** Aggregated and cleaned monitoring & product-test datasets; standardized units/timestamps/site IDs; created a reproducible **Excel/Power Query** ETL workflow.
- **Exploratory data analysis (EDA):** Built PivotTables/charts and ad-hoc dashboards for trends, seasonality, and site variance; documented data dictionaries and assumptions.
- **Modeling & comparison:** Ran **linear/multiple regression** variants to compare a prototype product against a commercial baseline; reported **R² / adjusted R²**, RMSE/MAE, and sensitivity to covariates.
- **Quality control:** Outlier rules, missing-data handling, and provenance logs; sanity checks against historical NYS trends.
- **Deliverables:** Decision-ready tables/figures and a concise memo summarizing variant performance and limitations for internal stakeholders.

Peer Teacher - CNSE (College of Nanoscale Science and Engineering)

University at Albany, Albany, NY, USA

September 2024-December 2024

- **Tutored CNSE students 1:1** and in small groups on core CS/engineering coursework, focusing on concept mastery and problem-solving.
- **Led systematic debugging** support by reproducing issues, isolating root causes, and validating fixes with test cases.
- **Reinforced best practices** in code quality (clean structure, readability, documentation) and improved student outcomes on assignments/projects.

EDUCATION AND CERTIFICATIONS:

- **B.S. Computer Science - University at Albany** Jan 2023-Dec 2026
Albany, USA
- **High School Diploma, Experimental Sciences** Sep 2017-June 2021
National Organization for Development of Exceptional Talents (Sampad)
Ahwaz, Iran

CERTIFICATIONS:

- **Tutor Essentials Certificate - Purdue University** 2024

- **SUNY Start-up Certificate** 2024
- **Developing AI Applications with Python and Flask** 2023
- **Python for Data Science, AI & Development** 2023
- **Django Application Development with SQL and Databases** 2023

ACHIEVEMENTS:

- Lita & Stephen Greenwald '40 Fund Award - Recognized for outstanding research contributions.** 2024
- Dean's List for Distinguished Students, University at Albany - Honored for academic excellence.** 2023
- Global Excellence Award Scholarship, University at Albany - Recognized for academic excellence.** 2023

RESEARCH & PROJECTS:

In Progress

Behavioral Profiling from Human-Computer Interaction (HCI Biometrics) - In progress, 2025-Present

Authors: Jari, R.; Dumas, C.

- **Signals:** keystroke dynamics, mouse trajectories, window-focus telemetry (privacy-aware logging).
- **Features:** dwell/flight times, n-gram intervals, velocity/accel bursts, path curvature, idle micro-pauses.
- **Models:** classical (LR, SVM, RF) vs. sequence models (RNN/TCN/Transformer); one-class/autoencoder anomaly scoring.
- **Eval:** ROC-AUC / EER, DET curves, **cross-device** and **cross-session** robustness; ablations on sampling rate, noise, and feature subsets.
- **Governance:** on-device hashing, minimal retention, opt-in de-identification; reproducible notebooks and metric dashboards.

Transformative Learning Analytics Initiative (TLAI): Tutor Insight & Development - In progress, 2025-Present

Authors: Dumas, C.; Jari, R.

- **Capture:** lightweight session signals (talk-time ratio, question-type mix, **Bloom level**, time-to-clarity).
- **Indicators:** interpretable formative metrics (not prescriptive scoring) with **fairness checks** across courses/cohorts.
- **Reliability & validity:** inter-rater agreement, session-to-session stability, correlations with learner progress proxies.
- **Feedback loops:** privacy-preserving dashboards that surface growth areas over time; export to pandas/SPSS/SAS.
- **Ethics:** anonymization, RBAC, IRB-aligned workflows; documentation for program governance.

VR Library Crisis Training with GenAI - IRB approved; ASIS&T 2025 accepted; published in JASIST 2026

Investigators: Dumas, C.; Williams, R. D.; Ogden, L.; Porwol, L.; Jari, R.; Borji, S.; et al.

- Built LLM-driven NPC dialogue with **STT** → **LLM** → **TTS** voice loop; **queued evaluator** to maintain VR FPS with retries/fallbacks.
- Logged HCI/learning metrics; anonymized exports to Python/SPSS; figures and methods for manuscripts.
- Usability and efficacy study design; noise/latency profiling; reproducible analysis pipeline.

Early Sjögren's Syndrome Screening from Saliva via ATR-FTIR + ML - *In preparation, 2024-Present*

Authors: **Lednev, I. K.; Vyas, B.; Jari, R.**

- **Scope:** End-to-end pipeline from raw ATR-FTIR saliva spectra to reproducible, leakage-safe classifiers for distinguishing Sjögren's-related classes vs. controls; iterative method development to maximize accuracy and robustness.
- **Preprocessing:** Baseline correction (e.g., AsLS/IModPoly-style), smoothing/denoising, vector/area normalization; per-sample scan aggregation; strict **subject/sample-wise** (group-based) splits to prevent leakage.
- **Features & Dimensionality:** Derivative spectra (1st/2nd), spectral band/region tuning, PCA/PLS component sweeps; systematic ablations over spectral ranges and preprocessing windows.
- **Modeling Strategy:** Progressed from classical baselines (PLS-DA/SVM-DA/logistic regression/random forest/k-NN/LDA; exploratory CNN) to **multi-class structured learning**, including **hierarchical 3-class classification** (N vs X, then R vs S) and **one-vs-one (OVO) 3-class** training, with reproducible run tracking.
- **Validation and Reporting:** Stratified CV / repeated runs with fixed seeds; confusion-matrix-driven analysis with balanced accuracy and macro-F1; ROC/AUC and clinically relevant sensitivity/specificity operating points where applicable; run-to-run stability tracking.
- **Robustness & QC:** Class-wise outlier screening, scan-level QA checks, class-imbalance handling, and strict separation of subjects across train/validation/test to ensure realistic generalization.
- **Engineering:** Batchable scripting + notebooks, automated artifact generation (predictions, metrics summaries, figures), publication-ready plotting pipeline; de-identified dataset handling with hashed IDs and controlled-access workflow.
- **Status:** Manuscript drafting underway; ongoing performance improvements via region tuning, component/parameter sweeps, and structured-model refinements (hierarchical/OVO), plus targeted robustness and leakage checks.

Publications:

Peer-Reviewed Journal Articles in Progress

Dumas, C., Williams, R. D., Ogden, L. & Porwol, L., Zhang, J, Borji, S. **Jari, R.**,. Evaluating the Usability of Using Serious Virtual Reality Games and GenAI for Library Science Graduate Students to Communicate with Patrons in Crisis. (Target: *Education and Information Technologies*)

Jari, R., Huang, J., & Goel, S. When Does Immersion Help? A Scoping Review of Extended Reality Job-Related Training for Neurodivergent People. (In preparation)

Peer-Reviewed Journal Articles

Dumas, C., Williams, R. D., Ogden, L., Flanagan, J., Porwol, L., Zhang, J., Borji, S., & **Jari, R.** (2026). Evaluating the Usability of Using Virtual Reality for Library Science Graduate Students to Communicate with Patrons in Crisis. *The Journal of the Association for Information Science and Technology* (JASIST).

Peer-Reviewed Workshop Papers

Steinfeld, N., Dumas, C., Porwol, L., & **Jari, R.** (2026). Exploring Intergroup Dynamics in Multi-User Social Virtual Reality Spaces using AI-driven Analytics. *SAXR 2026 (Shaping Future Human Connection: Social Augmentation through XR Technologies, 1st Workshop edition at CHI Conference on Human Factors in Computing Systems)*. Barcelona, Spain, April 13-17, 2026.

PRESENTATIONS (LECTURES, PANELS, WORKSHOPS, POSTER SESSIONS):

Refereed Conference Presentations

Williams, R., Dumas, C., Zhang, J, Borji, S. **Jari, R.**, Nasierowski, N, & Stark, T. (2026) Smart Training for Tough Moments: A VR/GenAI Tool for Public Library Staff. *Public Library Association Conference*. Minneapolis, MN, 1-3, April 2026.

Steinfeld, N., Dumas, C., Porwol, L., & **Jari, R.** (2026). Exploring Intergroup Dynamics in Multi-User Social Virtual Reality Spaces using AI-driven Analytics. *SAXR 2026 (Shaping Future Human Connection: Social Augmentation through XR Technologies – 1st Workshop edition at CHI Conference on Human Factors in Computing Systems)*. Barcelona, Spain, April 13-17, 2026.

Dumas, C., Williams, R., Zhang, J, Borji, S. **Jari, R.**, & Nasierowski, N. (2025). Navigating Tensions Using Serious Games: Integrating VR, Gamification, and GenAI for De-Escalating Patron Crises in Libraries. *88th Annual Meeting Association for Information Science & Technology*. Washington DC, 14-18, November, 2025.

Refereed Conference Poster Presentations

Dumas, C., Williams, J, Borji, S. **Jari, R.**, Zhang J. & Stark, T. (2025). Development of an AI chatbot for VR Training for Patron Interactions. *88th Annual Meeting Association for Information Science & Technology*. Washington DC, 14-18, November, 2025.